



THE WAY FORWARD

PROJECT FINANCING

GEMALINK2A – GMD



2027 – 2046

GROUP 01
TCH426E

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EXECUTIVE SUMMARY

Figure 1: Summary

RECOMMENDATION: INVEST/LEND

NPV - TIP (mil VND)	7,014,869
IRR - TIP	20%
Payback (year)	7.69
NPV - EPV (mil VND)	1,237,304
IRR - EPV	28.10%
Payback (year)	8.70

Source: Team analysis

Figure 2: Scenario Analysis Result

Source: Team analysis

	Optimistic	Base	Pessimistic
NPV TIP (Billion VND)	11,206	7,015	3,750
IRR TIP	23%	20%	17%
Payback (TIP)	6.3	7.7	9.7
NPV EPV (Billion VND)	2,359	1,003	(239)
IRR EPV	33%	24%	15%
Payback (EPV)	8.55	12.2	-

Figure 3: Vietnam Seaport Cargo Throughput (2015-2024)

Source: DNSE

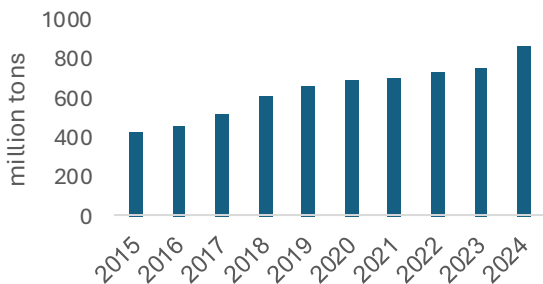


Figure 4: GMD Price Movements



Source: Yahoo!Finance

We recommend investing in the Gemalink 2A project because it offers a highly attractive risk-return profile with a sustainable financial structure and tight protection mechanisms for both lenders and investors. The project’s success is underpinned by the proven operational excellence of its first phase and the significant economic value it is expected to create above its cost of capital. Given the port's position as a premier international gateway and its robust debt-servicing capacity, this expansion is essential for capturing regional growth and maximizing long-term shareholder returns.

GEMADEPT'S PERFORMANCE SUPPORTS GEMALINK EXPANSION

Gemadept is Vietnam’s largest integrated port and logistics ecosystem, with port operations contributing 86.9% of its total revenue. In 2024, the group achieved 25.6% revenue growth and maintained a 14.7% national container market share. All GMD ports currently operate beyond their designed capacities with most notably is Gemalink Phase 1 at 110% utilization. This underscores the ability of generate great profit of the firm with great operational expertise which support the urgent necessity for the Phase 2A expansion.

STRONG EXPORT GROWTH AND GOVERNMENT INFRASTRUCTURE PLANS SUPPORT PORT DEMAND

The Vietnamese seaport industry benefits from an export-driven manufacturing sector and rapid integration into global supply chains. Demand is bolstered by a continuous annual GDP growth and the "China Plus One" strategy, which further signifies multinational manufacturing shifts to Vietnam. Furthermore, government master plans targeting on increased throughput by 2030 prioritize infrastructure investment in gateway clusters such as Cai Mep – Thi Vai further enhance the potential of Gemalink expansion.

ROBUST PROJECT ECONOMICS UNDER BASE-CASE SCENARIO

Under the base-case scenario, the project demonstrates robust financial viability from both the total investment and equity perspectives. From a Total Investment Perspective (TIP), the project generates a Net Present Value (NPV-TIP) and an Internal Rate of Return (IRR-TIP) exceeding the 8% weighted average cost of capital. From an Equity Perspective (EPV), the analysis also reveals an NPV-EPV and an IRR-EPV above the required 17% cost of equity. This suggests a safe zone for both investors and lenders of the project.

Figure 4 :GMD’s Port Ecosystem



Source: Gemadept

ABOUT GEMADEPT’S JOINT STOCK COMPANY

Gemadept Corporation has proven its ability to generate great cash flows, operate high-capacity ports efficiently and maintain a strong market position in Vietnam’s maritime sector. The exceptional performance of Gemalink Port, combined with the company’s integrated logistics network, underscores its necessity and ability to execute the Gemalink expansion project to further meet growing demand, and strengthen its competitive edge.

Business Overview: Gemadept Corporation (GMD), formerly a state-owned enterprise established in 1990, has affirmed its position as one of the pioneering and most important entities in the maritime and logistics industry in Vietnam. Throughout more than three decades of development, Gemadept has undergone a powerful transformation from a small freight forwarding agency into a multi-sector corporation with the largest integrated port and logistics ecosystem among listed companies on the Vietnamese stock market.

Development History: After equitizing in 1993 and listing on the Ho Chi Minh City Stock Exchange (HOSE) in 2002, the company expanded its investment into various fields such as rubber, real estate, and aviation services. However, from 2018, the leadership decided to focus on two main pillars, which are port operations and logistics. In that direction, by the end of 2024, Gemadept has owned and operated a network of seaports stretching from North to South, including deep-water ports, river ports, and Inland Container Depots (ICD).

REVENUE BREAKDOWN

The port operation segment serves as the core growth engine, contributing approximately 86.9% of the group’s total revenue in 2024. The Compound Annual Growth Rate (CAGR) of port operation revenue for the 2018-2024 period reached 9.1% according to MBS Research. This reflects the company’s ability to increase capacity and optimize their service fees including warehousing, customs inspection, and logistics services at the berth to generate good profit.

The logistics and related services segment contributes the remaining 13.1% of revenue. Although the revenue proportion tends to decrease compared to the previous period due to the strategy of focusing on ports, the logistics segment still plays an important role in attracting cargo to the group’s port system.

Profits from joint ventures and associates serve as a cornerstone of Gemadept Group’s earnings structure. In 2024, this income statement line item witnessed a staggering 197.4% year-on-year growth, reaching 815.6 billion VND. This exceptional performance was primarily driven by the unexpected results of Gemalink Port Phase 1, which contributed approximately 61.2% of the profit, at 548 billion VND within a year. This explosive growth and high profitability not only stressed the growing importance of this item but also showed the significant potential of Gemalink port.

FINANCIAL PERFORMANCE

Financial highlights in 2024 marked a successful year for Gemadept in terms of net revenue, showing a growth of 25.6% compared to 2023. This growth was mainly driven by the strong recovery of export orders to the US and Europe, along with the application of the new service price framework under Circular 39/2023/TT-BGTVT, which effectively bolstered the group’s pricing power and core operating margins.

Despite impressive revenue growth, the Net Profit After Tax of the parent company’s shareholders in 2024 reached 1,441 billion VND, a decrease of about 35.6% compared to the very high base level of 2023. However, this decrease is technical. The 2023 results were inflated by an extraordinary profit of nearly 1,844 billion VND from port divestment. When these are normalized, the number demonstrates a significant upward trend, signaling a stable and strong earning from the company.

Figure 6: Gemadept’s Revenue Contribution 2021-2025 (billions VND)

Source: Gemadept

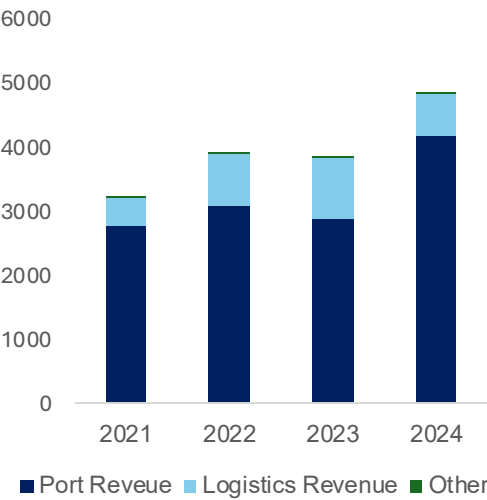


Figure 7: Gemadept’s Key Financial Indicators (billions VND)

Source: Gemadept

Financial Indicators	2023	2024
Net Revenue	3,845.83	4,832.02
Gross Profit	1,778.02	2,135.48
Net profit from Operation	3,177.03	2,418.58
Net profit after tax	2,533.93	1,923.58

Figure 7: Vietnam’s Port Ecosystem (2024)

Source: Vietnam Maritime Administration

Company	Market Share (%)
Saigon Newport Corporation	47%
Vietnam Maritime Corporation	20%
Gemadept Corporation	15%
Viconship	7%
Other smaller operators	11%

COMPETITIVE POSITION AND MARKET SHARES

In 2024, Gemadept Group solidified its 3rd position in the maritime industry by recording a total system-wide throughput of 4.4 million TEUs. This represents a 14.7% share of the national container market, which totaled 29.9 million TEUs nationwide according to the Vietnam Maritime Administration.

In 2024, all the company’s ports were operating over its designed capacity. These suggest a strong competitive edge in the area and bigger, in Vietnam’s port sector. The standout performer within this ecosystem is Gemalink Port. After only four years of operation starting from 2021 to 2024, Gemalink now handled 1.65 million TEUs, which is 110% of its designed capacity, being the second most contributors to the total real capacity of the company. Furthermore, it had the highest utilization rate among other ports in Cai Mep - Thi Vai area.

Figure 9: GMD's Port System Throughput Statistics (2024)

Source: Gemadept

Area	Seaport	Designed Capacity	Efficiency	2024's Capacity	Regional Market Share
North	Nam Dinh Vu, Nam Hai ICD	1.2 million TEUs/year	112%	1.36 million TEUs/year	19%
Central	Dung Quat	2 million tons/year	110%	2.2 million tons	34%
South	Phuoc Long ICD	500,000 TEUs/year	175%	1.4 million TEUs/year	28%
	Binh Duong	300,000 TEUs/year			
	Gemalink	1.5 million TEUs/year	110%	1.65 million TEUs/year	

Figure 10: Throughput Statistics of Seaport in Cai Mep – Thi Vai (2024)

Source: MBS analysis

Terminal	Capacity ('000 TEUs)	2023 Throughput ('000 TEUs)	2024 Throughput ('000 TEUs)	YoY Growth	Utilization Rate (2024)
TCIT	1,800	1,890	1,960	4%	109%
TCTT	1,000	820	1,090	33%	109%
CMIT	1,100	640	990	55%	90%
SSIT	1,500	500	680	36%	45%
GML	1,500	1,010	1,740	72%	116%
Total	6,900	4,860	6,460	33%	94%

Figure 10: GDP Growth rate of Vietnam (2020 – 2026F)

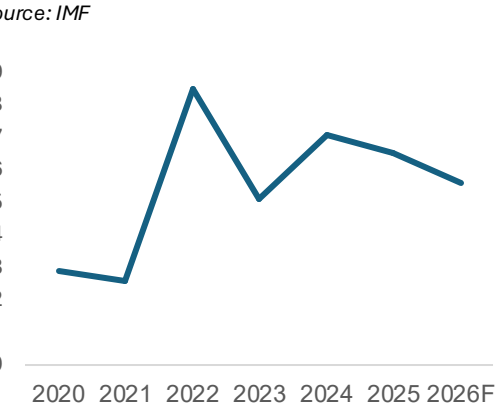
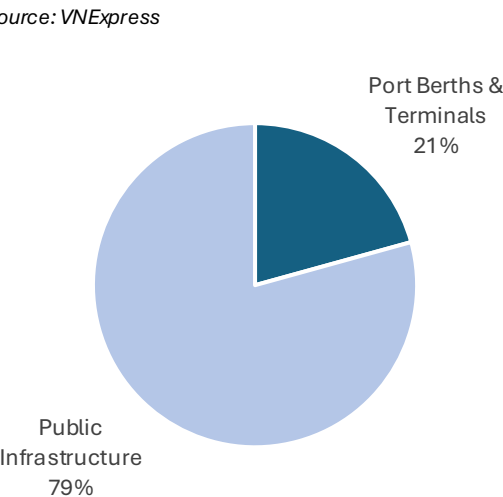


Figure 12: Government Infrastructure Investment Plan (2030)



OVERALL MARKET DYNAMICS – POSITIVE

Vietnam’s seaport industry benefits from the country’s highly open economic structure and strong export-oriented manufacturing sector. Furthermore, Vietnam’s deeper integration into global supply chains has also increased container shipping and strengthened the importance of major gateway ports. Although the sector faces short-term challenges such as congestion at key ports and rising logistics costs, the overall outlook remains positive.

NATIONAL ECONOMIC GROWTH SUPPORTING PORT DEMAND – POSITIVE

Vietnam’s economic expansion continues to support strong demand for seaport services. According to the IMF, Vietnam’s GDP is projected to grow at around 6-6.5% annually in the medium term, among the highest rates in Southeast Asia. In addition, Vietnam’s import-export turnover reached approximately USD 930 billion in 2025, while exports and imports together account for over 200% of GDP, reflecting the country’s highly trade-dependent economy. Major manufacturing hubs such as Binh Duong, Dong Nai, and Hai Phong generate significant cargo volumes that rely heavily on nearby deep-water port systems, according to VTC News in 2025.

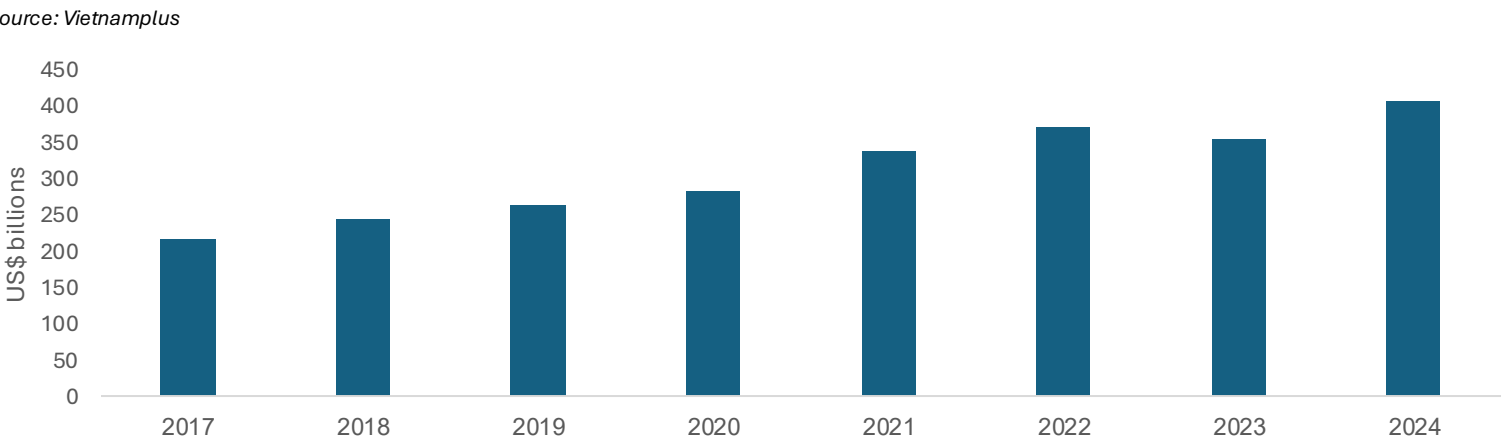
SUPPORTIVE REGULATORY FRAMEWORK AND PORT DEVELOPMENT – POSITIVE

The Vietnamese government is prioritizing the modernization of the national port system through long-term planning and infrastructure investment with 21% estimated seaport investment on Port Berths and Terminals by 2030. Furthermore, LuatVietnam website stated that decision 1579/QĐ-TTg approves the Seaport Development Master Plan for 2021–2030 with a vision to 2050, targeting total cargo throughput of 1.14 - 1.42 billion tons by 2030. This indicates potential infrastructure development opportunities and enhancement of global maritime trade for sea ports in general, alongside main port clusters such as Cai Mep - Thi Vai. Additionally, it allows Vietnam to accommodate ultra-large container vessels and maintain direct shipping connections with major markets such as North America and Europe.

GLOBAL SUPPLY CHAIN RELOCATION TREND – POSITIVE

Vietnam has become a key beneficiary of the global “China Plus One” strategy, as multinational companies shift manufacturing activities to Southeast Asia. Moreover, According to VietnamPlus, export-oriented industries such as electronics, textiles, and footwear continue to expand rapidly, supporting strong growth demand. The growth of these manufacturing sectors has increased export volumes and created strong demand for container shipping services. This trend generates additional cargo volumes and potential customers for container ports, particularly the Cai Mep – Thi Vai port cluster, which serves as a major gateway for export cargo from southern Vietnam’s industrial hubs.

Figure 12: Vietnam Merchandise Export Growth (2017–2024)



PROJECT DESCRIPTION

Figure 14: Gemalink's shareholders

Source: Gemadept

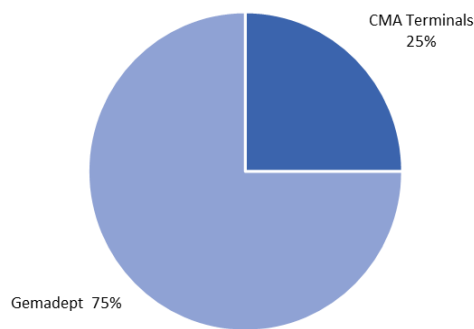


Figure 14: Key Assumptions

Source: Team analysis

No.	Item	Value	Unit
1	Project analysing life	20	years
2	Corporate income tax	20%	per year
3	Depreciation period		
	Basic infrastructure	30	years
	Machinery and equipment	15	years
4	Market value of machinery and equipment at disposal	20%	of original cost

Gemalink 2A is an expansion phase of Gemalink Deep-sea Port. The 33-hectare original Gemalink terminal, with capacity designed at about 1.5 million TEU per year, started construction in February 2019 and was officially put into operation in January 2021 has steadily established itself as a major gateway for international container traffic in Southeast Asia. The Phase 2A expansion aims to extend the terminal by approximately 11 hectares, increasing annual capacity by an additional 950,000 TEU in order to capture the growth of the industry.

OWNERSHIP AND SPV STRUCTURE

The Gemalink Deep-sea Port, including the Phase 2A expansion, is undertaken and operated by Gemadept - Terminal Link Cai Mep Joint Stock Company, which functions as the project’s special purpose vehicle (SPV). This SPV is jointly owned by Gemadept Corporation and CMA Terminals, the global terminal operating arm of the CMA CGM Group.

PROJECT TIMELINE

Phase 2A of Gemalink is part of a multi-stage development roadmap. Pre-construction planning and design work for Phase 2A have been progressing through 2024-2025, with formal construction scheduled to begin in the second quarter of 2025, and operations expected to commence in the third quarter of 2026. As of mid-2025, regulatory approvals and port licensing have been granted, and Gemadept has reaffirmed Phase 2A’s start timeline and operational expectations. However, information about the start of the construction has not been seen in 2025. The broader Gemalink master plan includes further downstream expansions of Phases 2B and beyond contingent, which was suggested to start when Phase 2A at 85% capacity.

PROJECT ECONOMICS & FINANCING STRUCTURE

Figure 15: Capital Disbursement Plan

Source: Team analysis

No.	Capital Disbursement Plan	2026				2027				Total
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	
1	Equity contribution	390,000	234,000	156,000	-	-	-	-	-	780,000
2	Bank loan	-	663,000	884,000	1,326,000	884,000	663,000	-	-	4,420,000
	Total	390,000	897,000	1,040,000	1,326,000	884,000	663,000	-	-	5,200,000
	Bank loan disbursement ratio		15%	20%	30%	20%	15%			
	Equity disbursement ratio	50%	30%	20%						

PROJECT ECONOMICS

Initial Investment and Capital Deployment

According to Gemadept, the total initial investment required for the project amounts to VND 5.2 trillion, including value-added tax. We allocated the investment to cover the development of port infrastructure, cargo-handling equipment, terminal facilities, and supporting logistics systems necessary for container terminal operations. The project is assumed to be implemented during the construction period from the second quarter of 2026 to the second quarter of 2027. During the construction, capital is disbursed in line with construction progress and equipment procurement schedules. The project’s capital assets will be depreciated using the straight-line depreciation method.

Investment Disbursement Schedule

Total equity contribution amounts to VND 780 billion, while bank loan financing accounts for VND 4.42 trillion, resulting in a capital structure of approximately 15% equity and 85% debt. Equity capital is injected first with 50% in Q1 2026 then gradually disbursed the rest in 2026. Bank debt is drawn afterward to support the main construction activities, with loan disbursements started first with 15% in Q2 2026 and ended in Q2 2027. This allocation is based on the project’s construction milestones and progress.

Figure 16: Operating Cost Assumptions

Source: Team analysis, OECD, ACCC

No.	Cost Item	Unit	Cost per TEU	Cost Structure (%)
I	Port operating costs			
1	Labor	USD	10.8	15%
2	Electricity	USD	3.6	5%
3	Fuel	USD	2.16	3%
4	Maintenance	USD	3.5	5%
5	Other expenses	USD	1.44	2%

Figure 17: Operating Expense Assumptions

Source: Team analysis, OECD, ACCC

No.	Item	Scale	Unit
1	Selling expenses	5.00%	% of revenue
2	General and administrative expenses	7.00%	% of revenue
3	Other expenses	1.20%	% of revenue
4	Maintenance CAPEX	2.00%	% of construction and equipment cost

Figure 18: Effective Capacity Assumptions

Source: Team analysis, OECD, ACCC

Effective Capacity year 1 (2027)	50%
Effective Capacity year 2 (2028)	65%
Effective Capacity year 3 (2029)	80%
Effective Capacity year 4 (2030)	90%
Effective Capacity year 5 (2031)	100%

Figure 20: Working Capital Assumptions

Source: Team analysis, OECD, ACCC

No.	Item	Ratio	Basis
1	Minimum cash balance	5%	Cash operating expenses
2	Accounts receivable	10%	Revenue
3	Inventory	0%	COGS
4	Accounts payable	8%	COGS
5	Working capital loan plan	50%	Working capital requirement
6	Working capital loan interest rate	8%	per year

PROJECT ECONOMICS

Revenue Assumptions and Price Growth

Revenue generation is driven by 20-foot container handling services, which depend on both throughput volumes and terminal tariffs in USD. The terminal is assumed to start operating at 50% of its installed capacity and generate its first profit in the first operating year of 2027, which consists of only Q4. The expected ramp-up phase is assumed to be 5 years, which is based on the industrial norms and to suit estimated exchange rate with annual price surge. The capacity will be set at maximum 100% regarding the information of the construction of Phase 2B. The base 20-foot container handling tariff is assumed at €72 per TEU, with an annual price and exchange rate escalation rates of 3% and 2% to enhance calculation of further analysis. This tariff, exchange rate and price surge assumptions are also benchmarked against pricing levels observed at comparable container terminals and historical information. The calculation of revenue is followed as below:

Throughput (TEU) = Effective capacity (%) × Design capacity (TEU)

Revenue (USD) = Throughput (TEU) × Handling fee (USD/TEU)

Revenue (VND) = Revenue (USD) × Exchange rate (USD/VND)

Operating Cost Structure and Expenses

The terminal’s operating cost structure is estimated based on standardized production costs per TEU handled by the port. The estimated cost amounts to approximately USD 21.5 per TEU, consisting of 5 key components. Furthermore, key expenses were also allocated by a percentage of revenue. These numbers are based on historical data of Gemalink Phase 1 and industrial benchmarks.

Working Capital Assumptions

The project will maintain a minimum cash balance equivalent to 5% of operating expenses to ensure adequate liquidity for daily operational needs. Accounts receivable are assumed to represent 10% of revenue, which is based on the typical credit period granted to shipping companies and logistics customers. Inventory is assumed to be 0%, as the container terminal operations are service-based. Meanwhile, accounts payable are estimated at 8% of cost of goods sold, which is also based on standard payment terms with suppliers and service providers. In addition, we assume that 50% of total working capital requirements can be financed through short-term working capital loans, allowing the project to maintain sufficient liquidity while reducing the need for additional equity funding.

Figure 20: Key Loan Assumptions

Source: Team analysis

No.	Item	Value	Unit
1	Loan tenor	15	years
2	Repayment period	12	years
3	Grace period	3	years
4	Interest rate	8%	per year

Figure 22: Equation for DSCR

Source: Principles of Project Finance

Debt Service = Interest + Principal

Actual DSCR = $\frac{CFADS}{Debt\ Service}$

Debt service = $\frac{CFADS}{Target\ DSCR}$

FINANCING STRUCTURE

Debt Financing Terms

The project assumes a long-term bank loan with an interest rate of 8%, which is based on the standard interest of infrastructure loan in Vietnam and the bank’s special offer to the company. The loan maturity is assumed to be 15 years with a 3-year grace period before principal repayment begins. The interest will be capitalized in the first 2 construction years. This structure allows the project to complete construction and establish stable operating cash flows before entering the full debt repayment phase.

Debt Repayment and Coverage Structure

Debt repayment follows a sculpted amortization schedule, which aligns annual debt service obligations with projected Cash Flow Available for Debt Service (CFADS). This structure is to maintain a stable Debt Service Coverage Ratio (DSCR) which will be suitable for the debt repayment ability of the company. The DSCR will be set at at least 1.2, which is the minimum DSCR for banks to accept the debt, and will be increased depending on CFADS performance of the company.

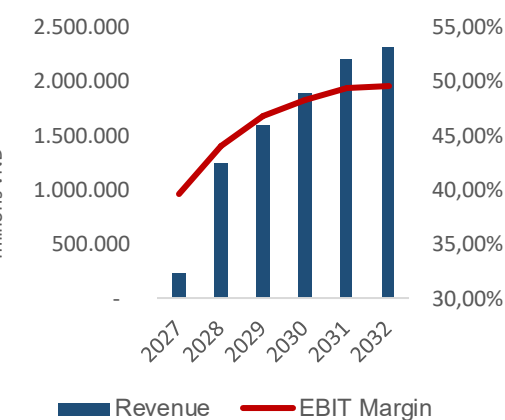
Lender Protection Mechanisms

To enhance financial resilience, the financing structure includes a Debt Service Reserve Account (DSRA) equivalent to six months of debt service, which is funded during the previous operational year before the debt obligations. Additionally, the investors are assumed to sign an Equity Support Agreement (ESA) to inject equity when the funding of DSRA and debt obligations after using DSRA cannot be fulfilled. This will enhance accuracy of the calculation and interpretation of later analysis.

OPERATING PERFORMANCE & CASH FLOW ANALYSIS

Figure 23: Revenue and EBIT Margin

Source: Team analysis



OPERATING CASH FLOW

We calculate the operating cash flow following this equation.

As we can interpret from the equation and economic assumptions above, the operating cash flow is affected mostly by depreciation. However, depreciation only accounts for a small amount of revenue, at highest around 17% at the first operating year and reduces significantly over time, reaching around 5% in 2046. This results in a continuous rise in operating cash flow of the project.

PORT REVENUE AND PROFITABILITY

Under the economic assumptions described above, the port’s revenue grows strongly during the initial ramp-up phase and stabilizes afterward. Revenue increases from VND 226,575 million in 2027 to VND 1,237,806 million in 2028 as operations begin, and continues growing at an annual rate of approximately 5.06% once the terminal reaches its steady operating capacity.

Profitability indicators remain strong throughout the project life. The model shows a gross profit margin of approximately 70% and an EBITDA margin of around 57%. After accounting for depreciation, the EBIT margin gradually increases during the ramp-up period and stabilizes at approximately 50%, which is consistent with the profitability levels observed in efficient port operations.

Figure 24. Equations for OCF

Source: Principles of Project Finance

$CF = (S - C - D)(1 - T) + D - NWCInv$

Or

$CF = (S - C)(1 - T) + T \times D - NWCInv$

Where:

S: Sales

C: Cash operating expense

D: Depreciation charge

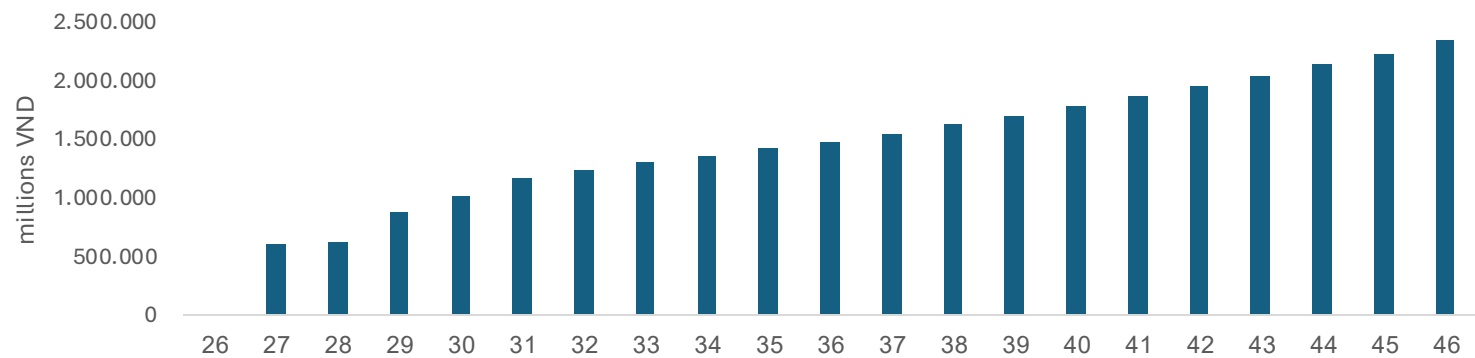
T: Tax rate

NWCInv: Investment (variation) in net working capital

OPERATING CASH FLOW

Figure 25: Operating cash flow

Source: Team analysis



As the project generates steadily increasing revenue grows and operating costs remain relatively stable as a proportion of revenue, operating cash flow rises significantly after the first operational year and continues to grow at roughly 4.5% every year when the port is stable. In conclusion, the project is able to produce consistent and expanding operating cash flows over the analysing period.

CASHFLOW WATERFALL

CFADS

The equation to calculate CFADS according to the book Principles of Project Finance is as followed:

$$\text{Cashflow before Funding} = \text{Revenue} - \text{Operating Expenditure(Opex)} - \text{Capital Expenditure(Capex)}$$
$$\text{CFADS} = \text{Cashflow before Funding} - \text{Debt\&Equity Funding} - \text{Corporate Income Tax} - \text{Working Capital Adjustments}$$

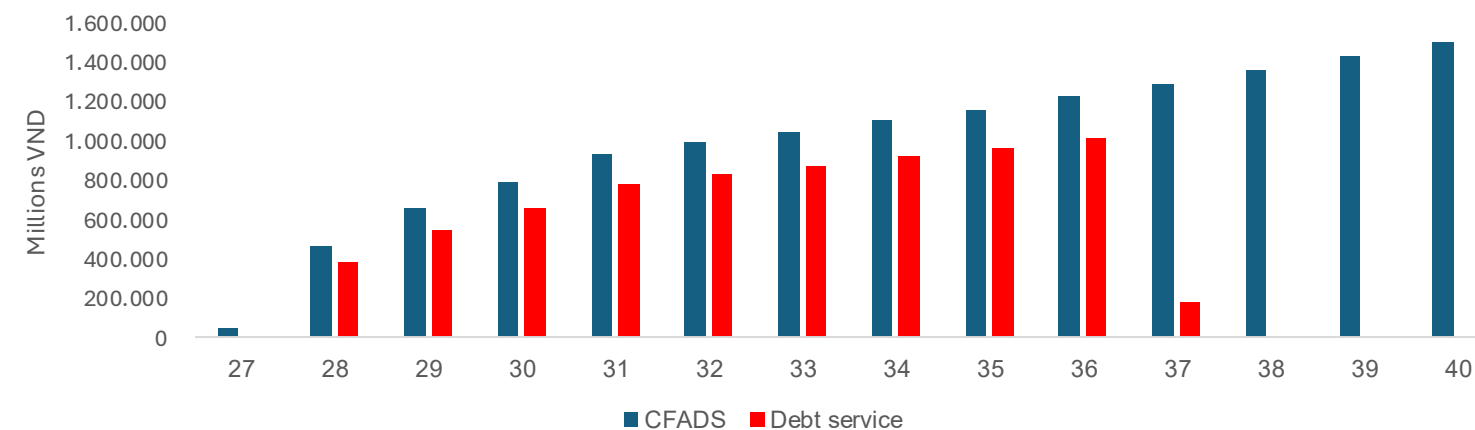
After adjusting the formula with our financing assumptions, the CFADS of the project is calculated as followed:

$$\text{Cashflow before Funding} = \text{Revenue} - \text{Cash Operating Expenditure} - \text{Maintenance Capital Expenditure}$$
$$\text{CFADS} = \text{Cashflow before Funding} - \text{Corporate Income Tax} - \text{Working Capital Adjustments}$$

As the maintenance capital expenditure is equal to 72,300 millions VND every year, with the projected growing operating performance, CFADS shows a similar trend with a steady increase by around 5.2% overtime, providing sufficient liquidity to support the project’s financing structure. At minimum target DSCR of 1.2, the project can comfortably fulfill its obligations before the maturity year by 3 years. Furthermore, after debt repayment, the port still have extra cash available left for reserve account.

Figure 26: CFADS and Debt Service at DSCR 1.2

Source: Team analysis



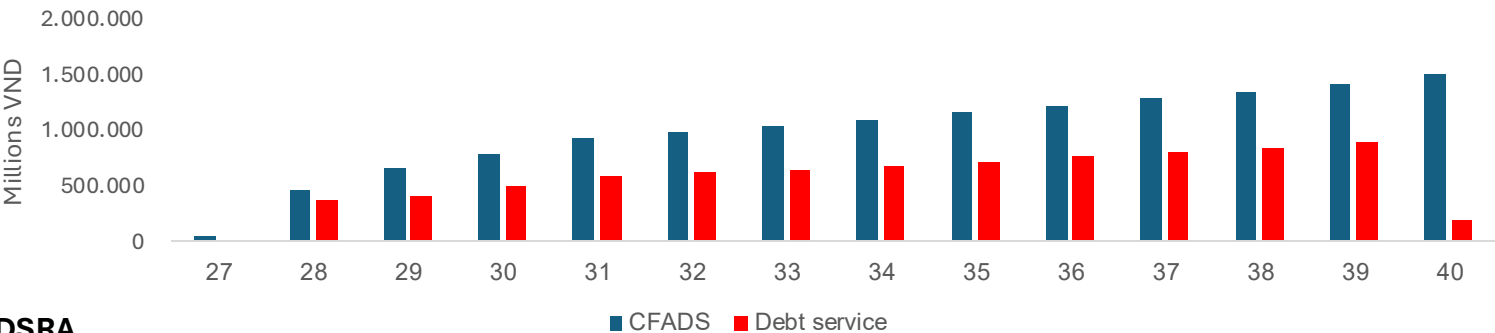
CASHFLOW WATERFALL

CFADS

We see that the maximum target DSCR for the first principal payment year is 1.71. Therefore, to provide additional comfort to lenders, the repayment profile was further structured so that the target DSCR increases to approximately 1.6, which is consistent with industry norms for, while still fulfilling debt obligations until maturity. Although in the 2027, the port cannot maintain the target DSCR of 1.6, the DSCR is still above 1.2 which signifies the ability of the port's debt repayment. At this rate of DSCR, the port also gains more extra cash available for reserve account.

Figure 25: CFADS and Debt Service at DSRC 1.6

Source: Team analysis



DSRA

After fulfilling debt service, the remaining available cash will be used to fund reserve account if needed. Noticeably, 2027 saw a lack of cash available for reserve account under both cases of DSCR, resulting in an equity injection from the investors. This is understandable as the port just started to operate for the first year. In the case of minimum DSCR of 1.2, the port also cannot fund its reserve account. This is caused by the high debt service that the port have to pay previously.

Figure 26: DSRA at DSCR 1.2

Source: Team analysis

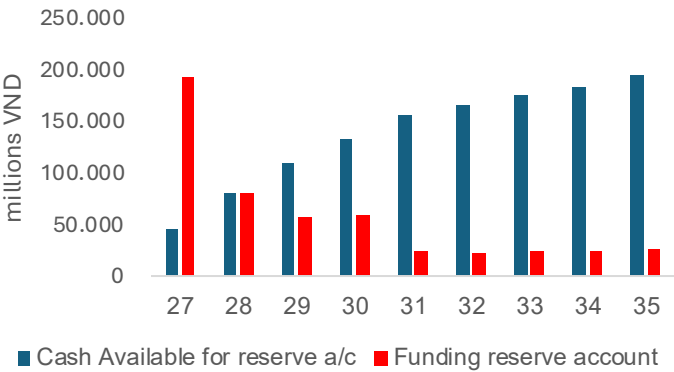
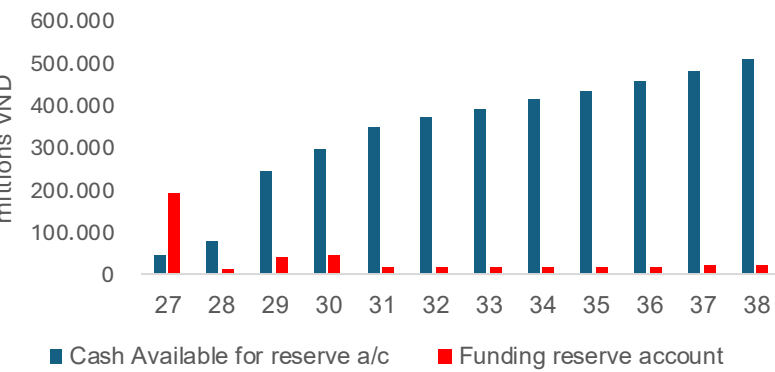


Figure 27: DSRA at DSCR 1.6

Source: Team analysis



Cash available for equity

The project cash available for equity is calculated followed the book Principle of Project Finance. As the project progresses and outstanding debt balances decline, a larger portion of the project's cash flow becomes available for equity holders, resulting in increasing returns to investors over time.

$$\text{Cash available for equity} = \text{CFADS} - \text{Debt Service} - \text{Reserve Account Funding} + \text{Reserve Account Releasing}$$

Figure 28: Cash available for equity

Source: Team analysis

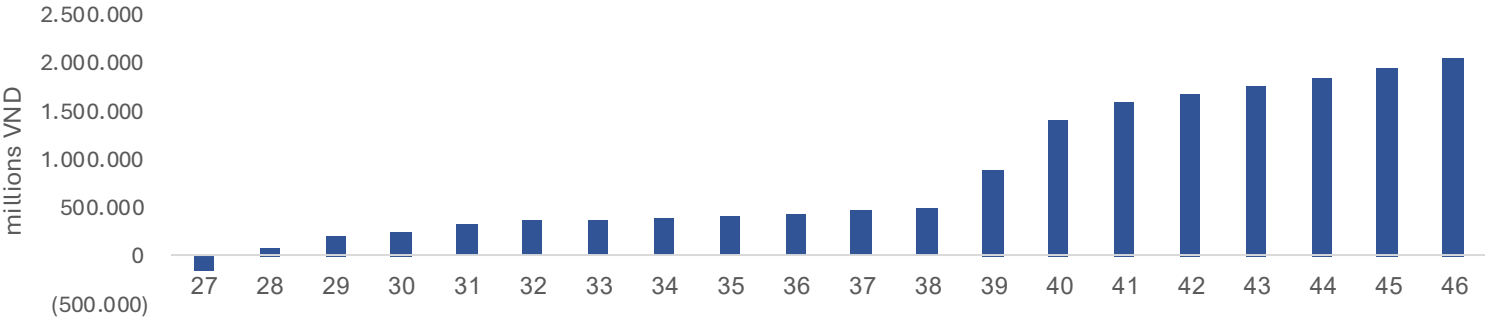


Figure 31: Actual DSCR

Source: Team analysis

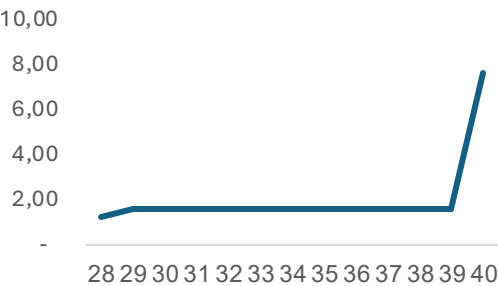
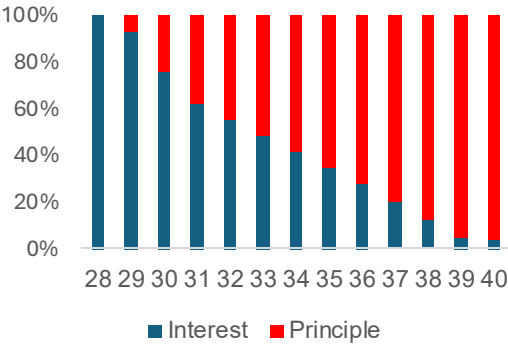


Figure 32: Debt Service

Source: Team analysis



Project Efficiency

Figure 33: WACC calculations

Risk-free rate	4.25%	10-year Vietnamese government bonds
Beta	1.5	Beta average of PHP, VSC, SGP and GMD
MRP	8.6%	Damodaran’s calculation on Vietnam’s emerging equity market
Cost of equity	17.1%	
Cost of debt	8.0%	Long-term lending rates for large infrastructure projects in Vietnam
Tax rate	20%	Current statutory corporate income tax rate of GMD
D/E	5.667	15% equity, 85% debt
WACC	8.00%	

For Cost of equity and WACC calculations we use the equations below:

Source: Team analysis

$$k_e = r_f + \beta(r_m - r_f)$$

Where:

k_e : Cost of Equity

r_f : Risk-free rate

β : Beta

r_m : Expected market return

$r_m - r_f$: Equity Risk Premium (ERP)

$$WACC = r_D \times (1 - T_c) \times \frac{D}{V} + r_E \times \frac{E}{V}$$

Where:

r_D : Cost of debt

T_c : Corporate tax rate

D : Market value of debt

E : Market value of equity

V : Total firm value ($V = D + E$)

r_E : Cost of equity

We use two valuation perspectives are applied which are the TIP (TIP) and the Equity Perspective (EPV). These approaches provide complementary views of project viability by examining returns at both the overall project level and the equity investor level.

Under TIP, valuation is based on free cash flows calculated by subtracting OCF with capital expenditure. These cash flows represent free cash flow to the firm and are discounted at the WACC of 8% to estimate NPV. **Under EPV**, the analysis focuses on the residual cash flows available to equity investors calculated by using cashflow available for equity. These equity cash flows are then discounted at the cost of equity of 17%, reflecting the higher risk borne by shareholders.

We conducted several analyses to evaluate the financial viability and risk profile of the project. Overall, the results indicate that the project is financially viable and bankable for lenders and also generates positive value for investors. Although equity returns remain sensitive due to the leveraged financing structure, the project still presents a balanced risk-return profile for both lenders and investors under the base-case assumptions.

Actual DSCR and LLCR

Debt Service Coverage Performance

The model shows an actual DSCR of 1.21 in 2028, which reflects the initial operational phase when the project begins generating revenue but cash flows have not yet fully reached the required DSCR but have passed the minimum DSCR. From 2029 onward, the DSCR stabilizes at the target level of 1.6. This indicates that the port has been able to align with the required debt payment. At the end of the loan tenor, DSCR increases to approximately 7.657. This suggests a strong improvement in the project’s debt servicing capacity over time, which indicates a low chance of default for lenders.

Long-Term Debt Coverage Metric

The project’s long-term financial strength is further assessed through the Loan Life Coverage Ratio (LLCR). The model estimates an LLCR of 1.63. This indicates that the present value of projected cash flows during the loan period is 1.63 times the outstanding debt balance. This is in range with the typical industry range of 1.40-1.60 for infrastructure financing according to Oxford. This indicates a strong capacity to sustain debt repayment over the years and provides a comfortable buffer for lenders.

Figure 34: Cashflow NPV - EPV

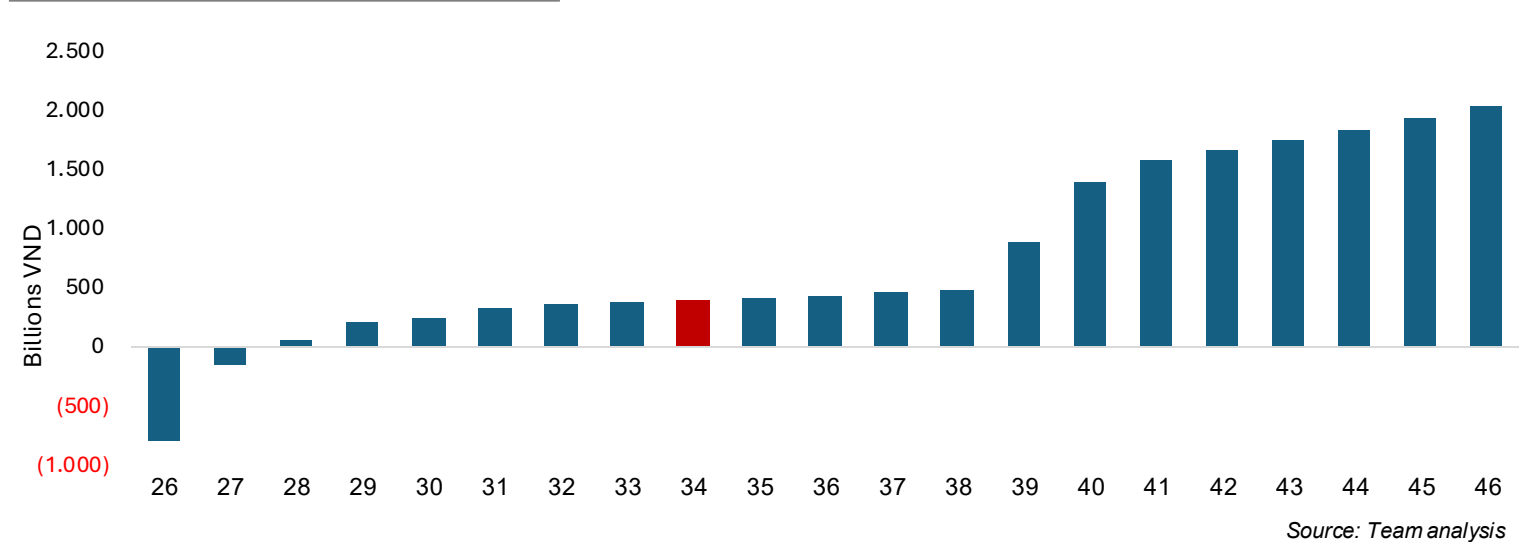
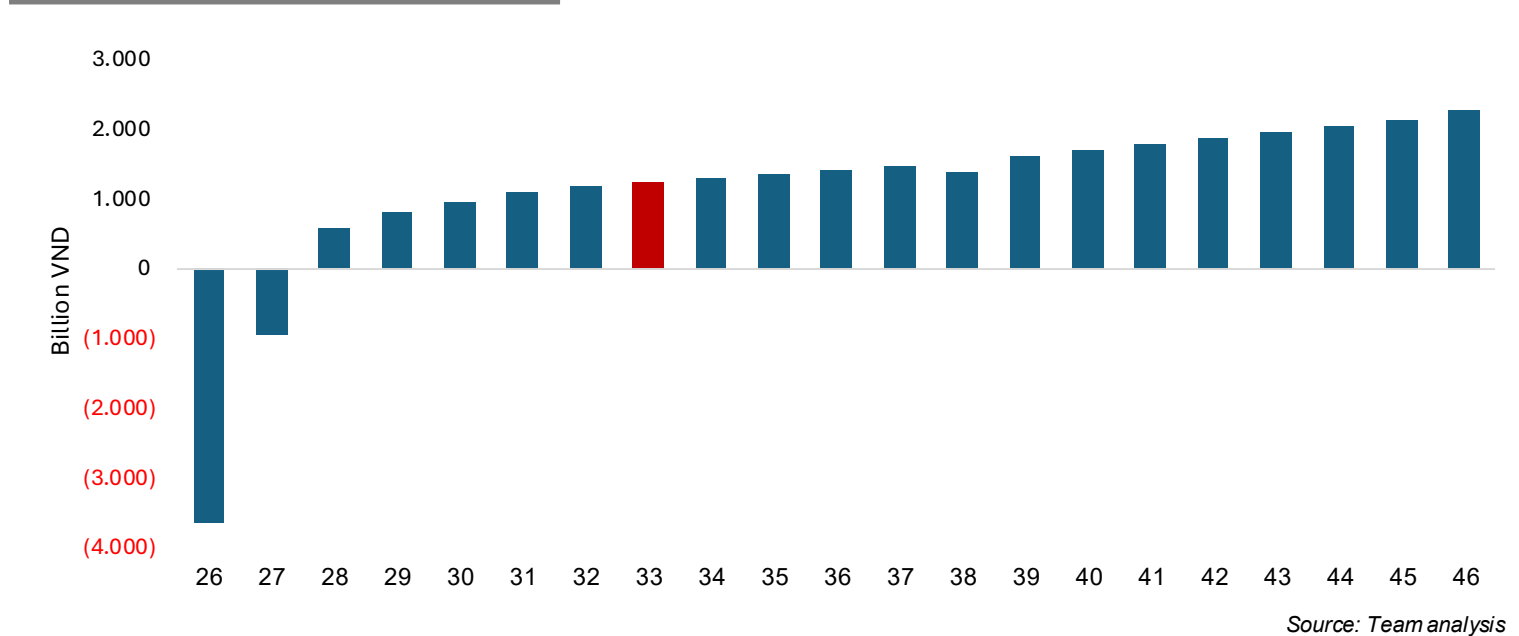


Figure 35: Cashflow NPV - TIP



Total investment perspective (TIP): the project generates an NPV-TIP of VND 7,014,869 million and an IRR-TIP of 20%, which is significantly higher than the WACC of 8%. This indicates that the project is expected to create substantial economic value above its overall cost of capital. The project has a discounted payback period of approximately 7.69 years, suggesting that the initial investment can be recovered within a relatively fast timeframe.

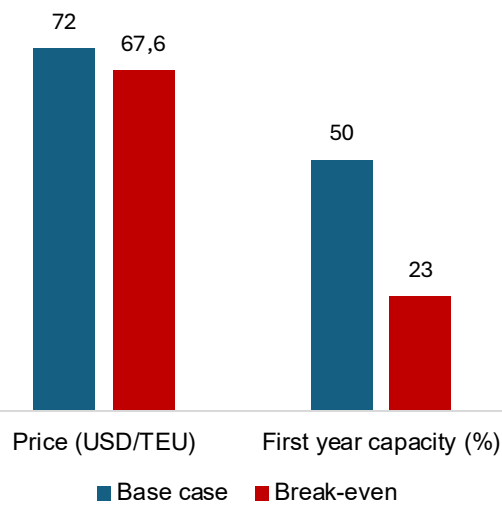
Equity perspective (EPV): the project generates an NPV-EPV of VND 1,276,304 million and an IRR-EPV of 28%, which exceeds the required cost of equity (Ke) of 17%. This indicates that the project is capable of generating returns above the rate required by equity investors, thereby creating positive value for shareholders. However, due to the project’s leveraged financing structure and the priority of debt servicing in the cash flow waterfall, the discounted payback period for equity investors is approximately 8.7 years.

Figure 36: Project efficiency

	NPV (billion VND)	IRR	Payback (year)
TIP	7,014	20%	7.69
EVP	1,237	28%	8.7

Source: Team analysis

Figure 37: Break-even



Source: Team analysis

Break-even analysis

The break-even analysis was conducted using Excel Solver. The analysis focuses on NPV-EPV = 0 while maintaining the minimum DSCR requirement of 1.2 and satisfies the payment of debt until maturity. This approach is adopted because NPV-TIP remains strongly positive across realistic operating ranges, whereas equity returns are more sensitive to changes in key operating drivers due to the project’s leveraged capital structure.

Two critical variables were selected for the Solver analysis, which are initial operating capacity and container handling tariff. These variables are chosen because they are the primary revenue drivers of the project.

The results show that the project reaches its equity break-even point when initial operating capacity declines to approximately 23%, and the minimum container handling tariff required to maintain financial viability is approximately USD 67.57 per 20-foot container. This is considerably below the base case capacity assumption of 50% and 72 USD. At this point, NPV-EPV equals zero, IRR-EPV equals the cost of equity of, and our DSCR constraint remains at the minimum required level of 1.2 and be able to pay fully their debt at the maturity date. However, the investors will have to continuously inject equity for 4 years, which shows the vulnerability of available cash flows.

Scenario analysis

Figure 38: Scenario assumptions

No.	Factors	Optimistic	Base	Pessimistic
1	Initial operating capacity from the first year	60%	50%	40%
2	Container handling tariff (20-foot container)	74.0	72.0	70.0
5	Price growth rate	4%	3%	2%
6	Accounts receivable	8%	10%	12%
7	Inventory	-	-	-
8	Accounts payable	10%	8%	6%
9	Long-term bank loan interest rate	6%	8%	10%
10	Production cost growth	-5%	0%	5%
11	Selling expenses	4.0%	5.0%	6.0%
12	Administrative expenses	6.0%	7.0%	8.0%
13	Other expenses	1.1%	1.2%	1.3%
14	Annual maintenance cost	1.5%	2.0%	2.5%

Source: Team complination

The analysis considers 11 key financial and operational drivers that directly affect the project’s revenue generation, operating efficiency, and financial sustainability. Additionally, we set the DSCR at minimum 1.2 for a fair comparison. The drivers as mentioned above include:

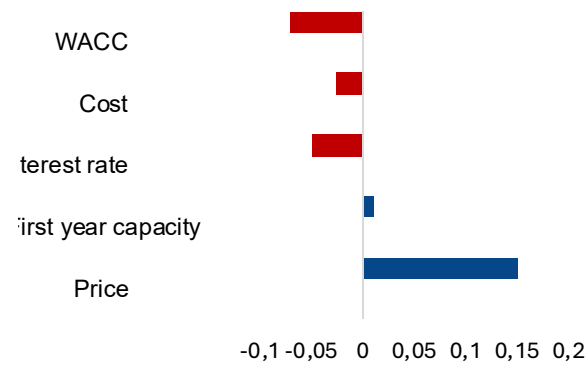
- (i) **Initial operating capacity.** Considering the industrial norm of 40–60% utilization in the initial years of ports, we minus 10% for the optimistic case and decrease 10% to the initial operating capacity in the pessimistic case.
- (ii) **Container handling tariff for 20-foot containers.** Considering container handling tariffs at deep-sea terminals in Vietnam generally range between 70–75 USD per 20-foot container. We set the range of 2 USD for other cases.
- (iii) **Annual price growth rate.** Considering approximately 3% increase in historical price, we set a range of 1% for cases.
- (iv) **Working capital** Considering **account receivable** equivalent to 8–12% of operating expense and for accounts payable is 10%-6%, based on industrial benchmarks, we set a 2% range for both accounts in pessimistic and optimistic cases.
- (v) **The long-term bank loan interest rate,** Considering borrowing cost for large infrastructure projects in Vietnam, consistent with the 7-10% lending range reported by major commercial banks such as Vietcombank, BIDV, and VietinBank, we select a range of 2% for the cases
- (vi) **Major operating cost** components considering the actual history percentage data of GMD and industrial benchmark, we consider a 5% change in the pessimistic and optimistic cases.

Figure 39: Scenario result

	Optimistic	Base	Pessimistic
NPV TIP (Billion VND)	11,206	7,015	3,750
IRR TIP	23%	20%	17%
Payback (TIP)	6.3	7.7	9.7
NPV EPV (Billion VND)	2,359	1,003	(239)
IRR EPV	33%	24%	15%
Payback (EPV)	8.55	12.2	-

Source: Team analysis

Figure 40: One factor sensitivity



Source: Team analysis

Sensitivity analysis

A one-factor sensitivity analysis was conducted to assess how changes in key variables affect the project’s Net Present Value (NPV) on base-case assumptions. Each variable was increased by 5%, while all other assumptions were held constant . The selected variables include the container handling price, first-year operating capacity, interest rate, operating cost, WACC. Moreover, we set the DSCR at minimum 1.2. for a fair calculation. These factors represent the key drivers affecting the project’s revenue generation, operating margin, and financing cost.

The results show that the **container handling price** has the most significant positive impact on the project’s value. A 5% increase in the handling price leads to a 15.1% increase in NPV, indicating that the project’s profitability is highly sensitive to tariff levels. However, a 5% increase in the WACC reduces NPV by 7.1%, making it the most influential negative factor in the analysis. This result highlights the importance of financing conditions, as higher capital costs significantly reduce the present value of future project cash flows.

As a result, we conducted a sensitivity analysis by varying the container handling price for 20-foot containers from –15% to +15%, while keeping all other assumptions constant. Although the result came out quite good for NPV-TIP and IRR-TIP even in pessimistic case, it was more severe for NPV-EPV and IRR-EPV when the handling price decreases by 15%, the equity NPV turns go low at VND 299 billion VND and IRR declines to 20%. This indicates that substantial price reductions could significantly weaken the project’s profitability and create downside risks for equity investors. This difference is mainly driven by the project’s highly leveraged financing structure.

Figure 41: Price change sensitivity

% change in price								
% change in price		-15%	-10%	-5%	0%	5%	10%	15%
NPV - TIP (million VND)	7,014,869	4,592,599	5,400,022	6,207,446	7,014,869	7,822,293	8,629,716	9,437,140
IRR - TIP	20%	17%	18%	19%	20%	21%	22%	23%
NPV - EPV(million VND)	1,276,049	299,129	639,398	959,157	1,276,049	1,599,295	1,930,678	2,267,791
IRR - EPV	29%	20%	24%	26%	29%	31%	33%	35%

Source: Team analysis

Scenario analysis

Optimistic Scenario

Under favorable market conditions, the project generates strong financial performance from both the NPV TIP and NPV EPV. The results show that the NPV TIP reaches 11,206,081, while the IRR is 23%, indicating that the project generates returns well above the required cost of capital. From the equity perspective, the project becomes even more attractive due to improved financing conditions, with an NPV of 2,359,044 and an IRR of 33%. Furthermore, the port is able to pay fully its debt before the maturity date of 5 years, with the maximum target DSCR for the first principal payment year is 2.916. This signals a safe zone even for lenders.

The project remains highly attractive at both firm and equity level, as the IRR of 23% and 33% is substantially higher than WACC, 8%, confirming strong value creation under favorable market conditions.

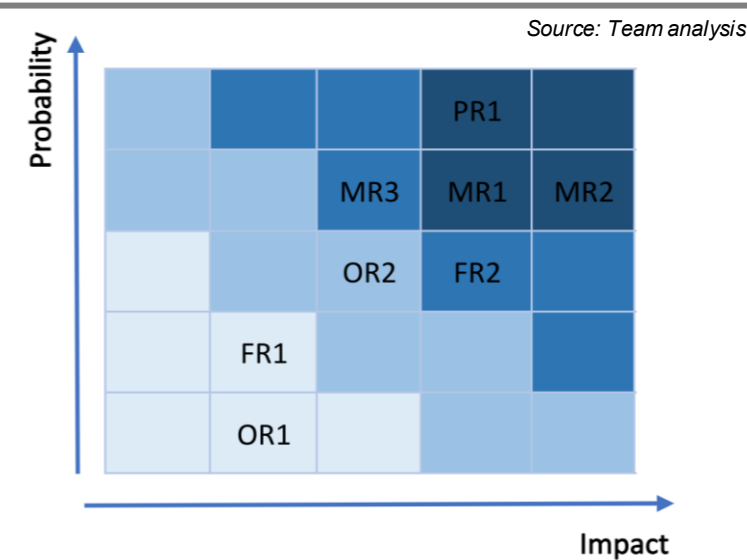
Pessimistic Scenario

Under these less favorable conditions, the project’s financial performance declines significantly due to lower revenues as well as higher operating and financing costs.

From the total investment perspective, the NPV decreases to 3,749,513, with an IRR of 17%, indicating that although the project still generates positive value, its profitability is substantially lower, less than half, compared to the base and optimistic scenarios.

From the equity perspective, the impact is more pronounced. The NPV becomes negative at (248,695) and the IRR falls to 15%, suggesting that equity investors face higher risk and reduced returns under adverse market conditions. With Ke at 17.1%, the project is not financially attractive under this downside scenario, as return generated for shareholders is insufficient to meet their required rate of return. Furthermore, the port cannot fulfill its debt obligation in 15 years, which suggests a dangerous default risk for lenders.

Figure 41: Risk Matrix



Operational Risk

[OR1] Delay in fund disbursement - Low risk

While this risk has a lower probability and impact compared to the others, delays in funding may temporarily constrain cash flow, affect CAPEX schedule and require the use of working capital or contingency funds. Gemalink 2A involves complex equipment installation and phased operational start-up, so even a short delay in fund release can affect early throughput and reduce initial revenue.

Mitigation: Gemalink prepares for funding delays by keeping a contingency fund (emergency cash) and a flexible working capital line. The project also uses a phased start-up plan, allowing it to begin operations in small steps.

[OR2] Operational & Equipment risk – Medium risk

This risk includes both higher operating costs (fuel, electricity, labor) and possible failures of important equipment, such as quay cranes, RTG cranes, and TOS software, which can reduce throughput and operational efficiency. While contingency funds and operational buffers can cover some of these problems, repeated or long-lasting equipment failures can cause ongoing revenue losses. Suppliers and maintenance contractors are affected, as equipment issues may increase their workload or put them at risk of penalties.

Mitigation: Gemalink uses tax breaks and fee waivers from Articles 15 & 16 of Law No.61/2020/QH14 on PPP to save money. This extra cash acts as a safety fund. If equipment breaks or power costs rise, the project can use these savings to fix problems quickly without affecting its daily spending or budget.

Market risk

[MR1] Actual throughput below plan - Extreme risk

Throughput risk happens when the actual number of containers handled is less than the planned capacity of 950,000 TEUs per year. The initial ramp-up is expected to reach 50%-65%, but operational problems or competition from nearby ports like Cai Mep Ha or Can Gio could reduce this further. Lower throughput directly reduces revenue and makes it harder for the project to meet debt payments. Port operators are most affected, since lower throughput can show inefficiencies or delays in terminal operations.

[MR2] Competition from other ports - Extreme risk

As related to the previous section, competition from nearby ports like Cai Mep Ha and Can Gio is a strategic risk for Gemalink 2A because it could reduce market share and the number of containers handled. Expansions by competitors may also shorten the time available to attract new clients, delaying planned cash flows. Higher competition may require more marketing or service improvements, which can increase costs. Continuous monitoring of competitors and actively retaining clients is therefore important to manage this risk.

Mitigation: The project is protected by a "safety promise" in Article 13 of Law No.61/2020/QH14 on PPP. This law ensures that even if the government changes rules or new ports create more competition, Gemalink’s original tax deals and rights stay the same. This keeps the project stable and keeps customers confident.

Figure 43: Competitor analysis

Source: Team analysis

No.	Criteria	Gemalink 2A	Cai Mep Ha Logistics center & port	Can Gio International transshipment port
1	Project status	Expected operation: 2025–2026	Phase 1 expected: 2028	Phase 1 expected: 2027
2	Total capital	5.200.000 million VND	50.800.000 million VND	113.500.000 million VND
3	Throughput capacity	Phase 1: 1.5 million TEUs/year Phase 2a: +950.000 TEUs/year	10.8 million TEUs/year	16.9 million TEUs/year (Upon 7-phase completion)
4	Berth length	Total combined length: 1.5km	7.5 km	7.2 km
5	Vessel deadweight	250,000 DWT	250,000 DWT (24,000 TEUs)	250,000 DWT (24,000 TEUs)
6	Capital structure	85% Debt / 15% Equity	85% Debt / 15% Equity	80% Debt/ 20% Equity

Market Risk

[MR3] Container handling price fluctuation - High risk

This risk reflects the combined impact of container fee changes and competition. Even small drops in handling fees or market share can significantly reduce cash inflows, affecting NPV, IRR, and overall financial stability. High debt increases the effect, which is why it is classified as extreme. Equity investors are affected, since lower fees or market share directly reduce revenue and returns. The project's financial plans assume fees will rise over time, and any difference from this assumption may require changes in operations or pricing.

Mitigation: Gemalink can mitigate this by signing long-term service contracts with major shipping lines to lock in stable prices. Additionally, the port can offer bundled services (like storage or logistics) to keep revenue high even if basic handling fees drop.

Financial risk

[FR1] Rising Interest rates - Low risk

If long-term market interest rates exceed the projected 8% per annum, debt service costs would rise, affecting cash flows and equity returns. The project's IRR provides some resilience and the interest rate is largely contractually fixed, reducing immediate concern; however, potential external factors, such as geopolitical volatility, could still influence market rates, requiring careful monitoring.

Mitigation: The project reduces this risk by using fixed-rate loan agreements, which keep interest costs at the planned 8% regardless of market changes. The project's strong IRR acts as a safety cushion to absorb small increases in debt costs.

[FR2] High debt / High D/E - High risk

Gemalink 2A's financing is heavily leveraged, with 85% debt and 15% equity. The combination of high debt, long-term interest payments, and large capital costs increases the financial risk. For this project, even small delays in collecting revenue or higher operating costs can put pressure on the Debt Service Coverage Ratio (DSCR) and lengthen the time needed to recover the investment. This risk is critical because the project's financial health and investor returns rely heavily on cash coming in on time and in full. Equity investors are especially affected, as any shortfall directly reduces the returns they receive.

Mitigation: The project also maintains a Debt Service Reserve Account (DSRA) — a special cash pot used only for debt payments. This ensures the port can always pay its bank loans even if revenue is temporarily delayed.

Political risk

[PR1] Political & Geopolitical risk - Extreme risk

Political risk, such as changes in trade policies, tariffs, or port regulations, can directly affect Gemalink 2A by reducing cargo volumes, increasing operating costs, and creating cash flow uncertainty, which may impact the project's debt repayment ability and profitability.

Take Vietnam in 2025 as an example, when the US imposed additional tariffs on Vietnamese exports, indicating a significant political and trade risk for Gemalink 2A. These tariffs raised the cost of Vietnamese goods in the US, which could reduce export volumes, especially for container shipments handled through the terminal.

The effects are several: first, throughput may fall as exporters lower shipment volumes or send goods through other ports or markets to avoid tariffs. This drop in container volumes directly reduces revenue and can make it harder for the project to meet debt payments.

Second, handling fees and profit margins may be affected if shipping lines renegotiate contracts, delay shipments, or pass higher costs back to ports by sending fewer containers. Third, tariffs may make cash flows less predictable, affecting financial indicators such as NPV and IRR, and possibly delaying the project's expected payback.

Mitigation: To handle US tariffs, Gemalink uses its freedom under Article 5 of Law No.61/2020/QH14 on PPP to change its business plan at any time. If trade with the US drops, the port can quickly sign new deals with ships from Europe or Asia.

The cargo volume destined for the US at Gemalink in 2024 and Q1/2025 accounted for less than 32%, with the remaining cargo being destined for Europe, Intra-Asia, etc. Gemalink also successfully attracted 4 new services routes to Africa, Europe, Canada and Brazil in April 2025 alone, significantly reducing US-bound cargo via Gemalink to approximately 20%.

APPENDIX 01: Capital distribution assumption

Base on industry norms and GMD actual financial statements report on previous projects

TT	Components	Weight	Note
1	Land	5%	Land clearance, site prep
2	Construction	30%	Civil + plant
	Bridge and deep water foundation	23%	Quay, marine works
	Facilities and road	8%	Access, yard surface
3	Building base	4%	Foundations
	Building foundation	3%	
	Euipment foundation	2%	
4	Equipment	35%	Core equipment
	STS cranes	13%	Main quay cranes
	RTG cranes (yard)	15%	Yard stacking
	Feeders / FCC cranes	4%	Feeder handling
	Haulage / Tractors	4%	Yard transport
	Support equipment	4%	Forklifts, spreaders
5	Design and installation	7%	IT + commissioning
	TOS / software	3%	Terminal Operating System
	Telecom & power install	2%	Cabling, power boards
	Commissioning	2%	Testing & handover
5	Contingency	6%	Cost overrun buffer
	Material price risk	3%	Price volatility
	Delay / other risk	3%	Construction risk
6	Working capital	4%	Startup working capital
	Operating cash buffer	2%	Initial Opex buffer
	Pre-COD logistics	2%	Fuel/crew/permits
7	Capitalised interest	9%	Interest during build
	TOTAL AFTER VAT	100%	

APPENDIX 02: Revenue calculation

No.	Item	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036
I	Gemalink Port 2A (million VND)	226,575	1,237,806	1,600,541	1,891,719	2,208,267	2,320,005	2,437,397	2,560,730	2,690,303	2,826,432
1	Design capacity (TEUs)	950,000	950,000	950,000	950,000	950,000	950,000	950,000	950,000	950,000	950,000
2	Effective capacity (%)	50%	65%	80%	90%	100%	100%	100%	100%	100%	100%
	Capacity change	0%	15%	15%	10%	10%	0%	0%	0%	0%	0%
3	Actual throughput (TEUs)	118,750	617,500	760,000	855,000	950,000	950,000	950,000	950,000	950,000	950,000
7	Tariff / Handling fee (USD per TEU)	72.00	74.16	76.38	78.68	81.04	83.47	85.97	88.55	91.21	93.94
8	Annual price escalation		3%	3%	3%	3%	3%	3%	3%	3%	3%
IV	Total production volume	118,750	617,500	760,000	855,000	950,000	950,000	950,000	950,000	950,000	950,000
VI	Total revenue (USD)	8,550,000	45,793,800	58,052,448	67,268,274	76,984,803	79,294,347	81,673,177	84,123,372	86,647,074	89,246,486
	Assumed exchange rate	26,500	27,030	27,571	28,122	28,684	29,258	29,843	30,440	31,049	31,670
VII	Revenue growth (%)	0%	436%	27%	16%	14%	3%	3%	3%	3%	3%

No.	Item	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046
I	Gemalink Port 2A (million VND)	2,969,449	3,119,704	3,277,561	3,443,405	3,617,641	3,800,694	3,993,009	4,195,056	4,407,325	4,630,336
1	Design capacity (TEUs)	950,000	950,000	950,000	950,000	950,000	950,000	950,000	950,000	950,000	950,000
2	Effective capacity (%)	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
	Capacity change	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
3	Actual throughput (TEUs)	950,000	950,000	950,000	950,000	950,000	950,000	950,000	950,000	950,000	950,000
7	Tariff / Handling fee (USD per TEU)	96.76	99.66	102.65	105.73	108.91	112.17	115.54	119.01	122.58	126.25
8	Annual price escalation	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%
IV	Total production volume	950,000	950,000	950,000	950,000	950,000	950,000	950,000	950,000	950,000	950,000
VI	Total revenue (USD)	91,923,880	94,681,597	97,522,045	100,447,706	103,461,137	106,564,971	109,761,920	113,054,778	116,446,421	119,939,814
	Assumed exchange rate	32,303	32,949	33,608	34,281	34,966	35,666	36,379	37,106	37,849	38,605
VII	Revenue growth (%)	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%

APPENDIX 03: Depreciation calculation

Items	Cost Q4/2027	D&A Q4/2027	Net FA Q4/2027	Range of life	Years left	Depreciation Rate	Annual Depr.
Plants	1,695,000	14,125	1,680,875	20-40	29.8	3.3%	56,500
Equipment	2,130,000	21,300	2,108,700	5-25	24.8	4.0%	85,200
Other Intangible Assets							
Software	150,000	3,750	146,250	3-10	9.8	10.0%	15,000

	2028	2029	2030	2031	2032	2033	2034	2035	2036
Depreciation Forecast for Tangible Assets	144,256	146,889	149,600	152,393	155,270	158,233	161,285	164,429	167,667
Depreciation forecast for intangible asset	15,000	15,000	15,000	15,000	15,000	15,000	15,000	15,000	15,000

	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046
Depreciation Forecast for Tangible Assets	171,002	174,437	177,975	181,619	185,373	189,239	193,221	197,323	201,547	204,757
Depreciation forecast for intangible asset	11,250	15,000	15,000	15,000	15,000	15,000	15,000	15,000	15,000	15,000

APPENDIX 04: COGS calculation

No	Item	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036
Table of COGS											
1	Labor	1,282,500	6,869,070	8,707,867	10,090,241	11,547,720	11,894,152	12,250,977	12,618,506	12,997,061	13,386,973
2	Electricit	427,500	2,289,690	2,902,622	3,363,414	3,849,240	3,964,717	4,083,659	4,206,169	4,332,354	4,462,324
3	Oil	256,500	1,373,814	1,741,573	2,018,048	2,309,544	2,378,830	2,450,195	2,523,701	2,599,412	2,677,395
4	Maintenance	415,625	2,226,088	2,821,994	3,269,986	3,742,317	3,854,586	3,970,224	4,089,331	4,212,011	4,338,371
5	Others	171,000	915,876	1,161,049	1,345,365	1,539,696	1,585,887	1,633,464	1,682,467	1,732,941	1,784,930
TOTAL (USD)		2,553,125	13,674,538	17,335,106	20,087,054	22,988,517	23,678,173	24,388,518	25,120,174	25,873,779	26,649,992
% growth			435.60%	26.77%	15.88%	14.44%	3.00%	3.00%	3.00%	3.00%	3.00%

No	Item	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046
Table of COGS											
1	Labor	13,788,582	14,202,240	14,628,307	15,067,156	15,519,171	15,984,746	16,464,288	16,958,217	17,466,963	17,990,972
2	Electricit	4,596,194	4,734,080	4,876,102	5,022,385	5,173,057	5,328,249	5,488,096	5,652,739	5,822,321	5,996,991
3	Oil	2,757,716	2,840,448	2,925,661	3,013,431	3,103,834	3,196,949	3,292,858	3,391,643	3,493,393	3,598,194
4	Maintenance	4,468,522	4,602,578	4,740,655	4,882,875	5,029,361	5,180,242	5,335,649	5,495,718	5,660,590	5,830,408
5	Others	1,838,478	1,893,632	1,950,441	2,008,954	2,069,223	2,131,299	2,195,238	2,261,096	2,328,928	2,398,796
TOTAL (USD)		27,449,492	28,272,977	29,121,166	29,994,801	30,894,645	31,821,484	32,776,129	33,759,413	34,772,195	35,815,361
% growth			3.00%	3.00%	3.00%	3.00%	3.00%	3.00%	3.00%	3.00%	3.00%

APPENDIX 05: WC needs calculation

No	Item	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036
	Revenue	226,575	1,237,806	1,600,541	1,891,719	2,208,267	2,320,005	2,437,397	2,560,730	2,690,303	2,826,432
	COGS	67,658	369,623	477,939	564,888	659,413	692,779	727,834	764,662	803,354	844,004
	Opex	97,566	533,013	689,211	814,595	950,904	999,020	1,049,570	1,102,679	1,158,474	1,217,093
1	Cash balance	4,878	26,651	34,461	40,730	47,545	49,951	52,479	55,134	57,924	60,855
	Change in cash balance	4,878	21,772	7,810	6,269	6,815	2,406	2,528	2,655	2,790	2,931
2	Receivables	22,658	123,781	160,054	189,172	220,827	232,001	243,740	256,073	269,030	282,643
	Change in receivables	22,658	101,123	36,273	29,118	31,655	11,174	11,739	12,333	12,957	13,613
3	Inventory	-	-	-	-	-	-	-	-	-	-
	Change in inventory	-	-	-	-	-	-	-	-	-	-
4	Payables	18,126	99,025	128,043	151,338	176,661	185,600	194,992	204,858	215,224	226,115
	Change in payable	18,126	80,899	29,019	23,294	25,324	8,939	9,391	9,867	10,366	10,890
5	Needs for working capital	9,410	51,407	66,471	78,564	91,711	96,351	101,226	106,349	111,730	117,383
6	working capital loan	4,705	25,703	33,236	39,282	45,855	48,176	50,613	53,174	55,865	58,692
7	ST interest	376	2,056	2,659	3,143	3,668	3,854	4,049	4,254	4,469	4,695

APPENDIX 05: WC needs calculation

No	Item	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046
	Revenue	2,969,449	3,119,704	3,277,561	3,443,405	3,617,641	3,800,694	3,993,009	4,195,056	4,407,325	4,630,336
	COGS	886,711	931,578	978,716	1,028,239	1,080,268	1,134,930	1,192,357	1,252,690	1,316,076	1,382,670
	Opex	1,278,678	1,343,379	1,411,354	1,482,769	1,557,797	1,636,621	1,719,434	1,806,438	1,897,843	1,993,874
1	Cash balance	63,934	67,169	70,568	74,138	77,890	81,831	85,972	90,322	94,892	99,694
	Change in cash balance	3,079	3,235	3,399	3,571	3,751	3,941	4,141	4,350	4,570	4,802
2	Receivables	296,945	311,970	327,756	344,341	361,764	380,069	399,301	419,506	440,733	463,034
	Change in receivables	14,302	15,025	15,786	16,584	17,424	18,305	19,232	20,205	21,227	22,301
3	Inventory	-	-	-	-	-	-	-	-	-	-
	Change in inventory	-	-	-	-	-	-	-	-	-	-
4	Payables	237,556	249,576	262,205	275,472	289,411	304,056	319,441	335,604	352,586	370,427
	Change in payable	11,441	12,020	12,629	13,268	13,939	14,644	15,385	16,164	16,982	17,841
5	Needs for working capital	123,323	129,563	136,119	143,007	150,243	157,845	165,832	174,223	183,039	192,300
6	working capital loan	61,661	64,782	68,059	71,503	75,121	78,922	82,916	87,111	91,519	96,150
7	ST interest	4,933	5,183	5,445	5,720	6,010	6,314	6,633	6,969	7,322	7,692

APPENDIX 06: Income statement

No	Item	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036
1	Revenuw	226,575	1,237,806	1,600,541	1,891,719	2,208,267	2,320,005	2,437,397	2,560,730	2,690,303	2,826,432
	%yoy	0.00%	446.31%	29.30%	18.19%	16.73%	5.06%	5.06%	5.06%	5.06%	5.06%
2	COGS	67,658	369,623	477,939	564,888	659,413	692,779	727,834	764,662	803,354	844,004
3	Depreciation	39,175	159,256	161,889	164,600	167,393	170,270	173,233	176,285	179,429	182,667
4	Management cost	15,860	86,646	112,038	132,420	154,579	162,400	170,618	179,251	188,321	197,850
5	Operation cost	11,329	61,890	80,027	94,586	110,413	116,000	121,870	128,036	134,515	141,322
6	Other cost	2,719	14,854	19,206	22,701	26,499	27,840	29,249	30,729	32,284	33,917
7	Financial expense	376	386,441	385,972	380,692	367,715	347,998	323,698	294,524	259,925	219,298
8	Earnings before tax	89,458	159,097	363,470	531,832	722,254	802,717	890,896	987,242	1,092,474	1,207,374
9	Tax	17,892	49,711	122,405	106,366	144,451	160,543	178,179	197,448	218,495	241,475
10	Net income	71,566	109,386	241,065	425,465	577,804	642,174	712,717	789,793	873,979	965,899

No	Item	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046
1	Revenuw	2,969,449	3,119,704	3,277,561	3,443,405	3,617,641	3,800,694	3,993,009	4,195,056	4,407,325	4,630,336
	%yoy	5.06%	5.06%	5.06%	5.06%	5.06%	5.06%	5.06%	5.06%	5.06%	5.06%
2	COGS	886,711	931,578	978,716	1,028,239	1,080,268	1,134,930	1,192,357	1,252,690	1,316,076	1,382,670
3	Depreciation	182,252	189,437	192,975	196,619	200,373	204,239	208,221	212,323	216,547	219,757
4	Management cost	207,861	218,379	229,429	241,038	253,235	266,049	279,511	293,654	308,513	324,124
5	Operation cost	148,472	155,985	163,878	172,170	180,882	190,035	199,650	209,753	220,366	231,517
6	Other cost	35,633	37,436	39,331	41,321	43,412	45,608	47,916	50,341	52,888	55,564
7	Financial expense	171,982	117,251	54,311	13,274	6,010	6,314	6,633	6,969	7,322	7,692
8	Earnings before tax	1,336,538	1,469,637	1,618,920	1,750,744	1,853,463	1,953,520	2,058,721	2,169,326	2,285,613	2,409,013
9	Tax	267,308	293,927	323,784	350,149	370,693	390,704	411,744	433,865	457,123	481,803
10	Net income	1,069,230	1,175,710	1,295,136	1,400,595	1,482,770	1,562,816	1,646,977	1,735,461	1,828,491	1,927,211

APPENDIX 07: Cashflow waterfall

No.	Objectives	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036
1	Revenue	226575	1237806.414	1600540.823	1891719.212	2208266.894	2320005.198	2437397.461	2560729.773	2690302.699	2826432.016
2	Opex	97565.7125	533013.1953	689210.6621	814595.3118	950904.2606	999020.0162	1049570.429	1102678.693	1158474.235	1217093.031
3	Capex	72300	72300	72300	72300	72300	72300	72300	72300	72300	72300
4	Cashflow before funding	56709.2875	632493.2187	839030.1607	1004823.9	1185062.633	1248685.182	1315527.032	1385751.08	1459528.465	1537038.985
6	Coporate income tax	11341.8575	126498.6437	167806.0321	200964.78	237012.5266	249737.0364	263105.4065	277150.216	291905.693	307407.797
7	Working capital adjustment	0	41997.00242	15064.56152	12092.80027	13146.40107	4640.553876	4875.365902	5122.059416	5381.235623	5653.526145
8	CFADS	45367.43	463997.5726	656159.5671	791766.3199	934903.7052	994307.5918	1047546.26	1103478.805	1162241.536	1223977.662
	Able to pay debt		TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
	Reserve: Withdrawal	0	0	0	0	0	0	0	0	0	0
	CFADS after reserve	45367.43	463997.5726	656159.5671	791766.3199	934903.7052	994307.5918	1047546.26	1103478.805	1162241.536	1223977.662
	Able to pay debt		TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
9	Debt service	0	384384.416	410099.7294	494853.9499	584314.8158	621442.2449	654716.4125	689674.2529	726400.9601	764986.0387
	Interest	0	384384.416	383312.9446	377549.2867	364046.362	344143.6813	319649.044	290270.1866	255456.235	214603.1296
	Principle	0	0	26786.78481	117304.6633	220268.4538	277298.5636	335067.3684	399404.0663	470944.7251	550382.9091
10	Cash Available for reserve a/c	45367.43	79613.15656	246059.8377	296912.37	350588.8895	372865.3469	392829.8475	413804.5518	435840.5761	458991.6232
	Able to reserve	FALSE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
	Reserve: Addition	192192.208	12857.65671	42377.11026	44730.43291	18563.71455	16637.0838	17478.92024	18363.3536	19292.53929	20268.74178
	Reserve: Releasing	0	0	0	0	0	0	0	0	0	0
11	Equity injection	146824.778	0	0	0	0	0	0	0	0	0
	Debt service	0	0	0	0	0	0	0	0	0	0
	Reserve a/c	146824.778	0	0	0	0	0	0	0	0	0
12	Cash available for equity	-146824.778	66755.49985	203682.7274	252181.9371	332025.1749	356228.2631	375350.9272	395441.1982	416548.0368	438722.8814

APPENDIX 07: Cashflow waterfall

2037	2038	2039	2040	2041	2042	2043	2044	2045	2046
2969449.5	3119703.6	3277560.623	3443405.19	3617641.493	3800694.152	3993009.276	4195055.546	4407325.356	4630336.019
1278677.9	1343379	1411354.021	1482768.535	1557796.623	1636621.132	1719434.161	1806437.53	1897843.269	1993874.138
72300	72300	72300	72300	72300	72300	72300	72300	72300	72300
1618471.5	1704024.6	1793906.601	1888336.655	1987544.87	2091773.02	2201275.115	2316318.016	2437182.088	2564161.881
323694.31	340804.92	358781.3202	377667.331	397508.974	418354.6041	440255.023	463263.6032	487436.4175	512832.3763
5939.5946	6240.1381	6555.889039	6887.617024	7236.130446	7602.278646	7986.953946	8391.093816	8815.683163	9261.756731
1288837.6	1356979.5	1428569.392	1503781.707	1582799.766	1665816.138	1753033.138	1844663.319	1940929.987	2042067.748
TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
0	0	0	0	0	0	0	0	0	0
1288837.6	1356979.5	1428569.392	1503781.707	1582799.766	1665816.138	1753033.138	1844663.319	1940929.987	2042067.748
TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
805523.52	848112.2	892855.8699	196393.299	0	0	0	0	0	0
167048.83	112068.07	48866.74197	7553.588424	0	0	0	0	0	0
638474.7	736044.13	843989.128	188839.7106	0	0	0	0	0	0
483314.11	508867.32	535713.522	1307388.408	1582799.766	1665816.138	1753033.138	1844663.319	1940929.987	2042067.748
TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
21294.34	22371.834	0	0	0	0	0	0	0	0
0	0	348231.2855	98196.64951	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
462019.77	486495.49	883944.8074	1405585.058	1582799.766	1665816.138	1753033.138	1844663.319	1940929.987	2042067.748

APPENDIX 08: Debt service at base case (DSCR = 1.6)

No.	Objectives	2026	2027	2028	2029	2030	2031
		1	2	3	4	5	6
1	Opening outstanding balance	-	2,959,190	4,804,805	4,804,805	4,778,018	4,660,714
2	Loan disbursement	2873000	1,547,000				
3	Principal repayment	-	-	-	26,787	117,305	220,268
4	Closing outstanding balance	2,873,000	4,506,190	4,804,805	4,778,018	4,660,714	4,440,445
5	Interest expense	86,190	298,615	384,384	383,313	377,549	364,046
6	Debt service (principal + interest payme	-	-	384,384	410,100	494,854	584,315
7	Interest during construction (IDC)		384,805				

2032	2033	2034	2035	2036	2037	2038	2039	2040
7	8	9	10	11	12	13	14	15
4,440,445	4,163,147	3,828,079	3,428,675	2,957,731	2,407,348	1,768,873	1,032,829	188,840
277,299	335,067	399,404	470,945	550,383	638,475	736,044	843,989	188,840
4,163,147	3,828,079	3,428,675	2,957,731	2,407,348	1,768,873	1,032,829	188,840	-
344,144	319,649	290,270	255,456	214,603	167,049	112,068	48,867	7,554
621,442	654,716	689,674	726,401	764,986	805,524	848,112	892,856	196,393

APPENDIX 09: REFERENCES

- An, V. (2026, January 5). *Imports and exports in 2025 are expected to reach a record high of over 930 billion USD*. VnEconomy.
<https://vneconomy.vn/xuat-nhap-khau-nam-2025-can-dich-ky-luc-hon-930-ty-usd.htm>
- LuatVietnam. (2021, September 22). *Decision 1579/QĐ-TTg 2021 master plan on Vietnam's seaport system development in 2021-2030*.
[https://english.luatvietnam.vn. https://english.luatvietnam.vn/decision-no-1579-qd-ttg-dated-september-22-2021-of-the-prime-minister-approving-the-overall-master-plan-on-vietnams-seaport-system-development-in-t-209873-doc1.html](https://english.luatvietnam.vn/https://english.luatvietnam.vn/decision-no-1579-qd-ttg-dated-september-22-2021-of-the-prime-minister-approving-the-overall-master-plan-on-vietnams-seaport-system-development-in-t-209873-doc1.html)
- Nguyet Duc. (2025, September 7). *By withdrawing from the rubber sector, Gemadept could earn 530 billion VND*. Báo Công Thương.
<https://congthuong.vn/rut-khoi-linh-vuc-cau-su-gemadept-co-the-thu-ve-530-ty-dong-419582.html>
- Phi Long. (2025, February 4). *Vietnam's seaports and maritime industry affirm their position on the international map*. Saigon Port.
<https://saigonport.vn/cang-bien-hang-hai-viet-nam-khang-dinh-vi-the-tren-ban-do-quoc-te/>
- Tap chi Cong Thuong. (2025, April 3). *Gemadept Group's (GMD) production output could exceed its designed capacity by 18% this year*.
<https://tapchicongthuong.vn/san-luong-khai-thac-cua-tap-doan-gemadept--gmd--co-the-vuot-18--cong-suat-thiet-ke-nam-nay-138992.htm>
- VietnamPlus. (2025, April 3). *Vietnam's trade shines as a pillar of economic growth*. En.vietnamplus.vn.
<https://en.vietnamplus.vn/vietnams-trade-shines-as-a-pillar-of-economic-growth-post312763.vnp>
- VNA. (2021, October 16). *Vietnam's seaports set to handle 1.14-1.42 billion tonnes of cargo by 2030*. Vietnam Seaports Association.
<https://www.vpa.org.vn/vietnams-seaports-set-handle-1-14-1-42-billion-tonnes-cargo-2030/>
- VTC News. (2025, November 4). *Cai Mep – Thi Vai Port Complex: A major gateway to the South, reaching international maritime standards*. Vietnam.vn.
<https://www.vietnam.vn/en/cum-cang-cai-mep-thi-vai-cua-ngo-lon-cua-mien-nam-vuon-tam-hang-hai-quoc-te>
- National Assembly. (2020, June 17). *Law on Investment (No. 61/2020/QH14)*. Thu Vien Phap Luat.
<https://m.thuvienphapluat.vn/van-ban/Doanh-nghiep/Luat-Dau-tu-so-61-2020-QH14-321051.aspx>
- Gemadept Corporation. (2025, April 8). *Announcement to Investors regarding Gemalink Port project and strategic operations*. Gemadept.
https://www.gemadept.com.vn/wp-content/uploads/2025/04/Thong-bao-gui-NDT_08.04.2025-ENG.pdf
- CafeF. (2025, January 30). *Ho Chi Minh City approves three enterprises to invest in Cai Mep Ha Port worth nearly 2 billion USD*. CafeF News.
<https://cafef.vn/tphcm-chap-thuan-3-doanh-nghiep-dau-tu-cang-cai-mep-ha-gan-2-ty-usd-188260130195619122.chn>
- Trieu Vu. (n.d.). *Key facts about the Can Gio Super-Port Project and its regional impact*. Trieu Vu Logistics Knowledge.
<https://trieuvu.com/kien-thuc-can-nam/nhung-dieu-can-biet-ve-sieu-du-an-cang-can-gio-3427.html>
- Gemadept. (2025). *IR Newsletter*.
<https://www.gemadept.com.vn/wp-content/uploads/2025/08/IR-NEWSLETTER-Q22025-.pdf>
- Gemadept Corporation. (2026, January 8). *Terminal Handling Charge at deep - sea ports to rise starting February 1, 2026* - Gemadept.
<https://www.gemadept.com.vn/en/terminal-handling-charge-at-deep-sea-ports-to-rise-starting-february-1-2026/>
- Vietnamnews.vn. (2024). *Việt Nam's port firms thrive in 2024*.
<https://vietnamnews.vn/economy/1691614/viet-nam-s-port-firms-thrive-in-2024.html>

APPENDIX 09: REFERENCES

- International Monetary Fund. (2024, October). *World economic outlook database*. International Monetary Fund.
<https://www.imf.org/en/Publications/WEO/weo-database>
- World Bank. (2023). *World development indicators*. World Bank.
<https://databank.worldbank.org/source/world-development-indicators>
- United Nations Conference on Trade and Development. (2023). *Review of maritime transport 2023*. United Nations.
<https://unctad.org/publication/review-maritime-transport-2023>
- World Bank. (2017). *Public-private partnerships reference guide* (3rd ed.). World Bank.
<https://ppp.worldbank.org/public-private-partnership/library/ppp-reference-guide-3rd-edition>
- Asian Development Bank. (2018). *Public-private partnership handbook*. Asian Development Bank.
<https://www.adb.org/documents/public-private-partnership-handbook>
- Organisation for Economic Co-operation and Development. (2014). *Infrastructure financing instruments and incentives*. OECD.
<https://www.oecd.org/finance/private-pensions/infrastructure-financing-instruments-and-incentives.htm>
- International Finance Corporation. (2012). *IFC infrastructure finance handbook*. International Finance Corporation.
<https://www.ifc.org>
- World Bank. (2019). *Vietnam logistics report: Improving logistics performance*. World Bank.
<https://documents.worldbank.org/en/publication/documents-reports/documentdetail/516301554451121430>
- Asian Development Bank. (2021). *Vietnam transport sector assessment, strategy, and road map*. Asian Development Bank.
<https://www.adb.org/documents/viet-nam-transport-sector-assessment-strategy-and-road-map>
- Vietnam Maritime Administration. (2023). *Vietnam seaport statistics report*. Vietnam Maritime Administration.
<https://vinamarine.gov.vn>
- Notteboom, T., Pallis, A., & Rodrigue, J. P. (2022). *Port economics, management and policy*. Routledge.
<https://porteconomicsmanagement.org>
- Stopford, M. (2009). *Maritime economics* (3rd ed.). Routledge.
<https://www.routledge.com/Maritime-Economics/Stopford/p/book/9780415275583>
- Talley, W. K. (2009). *Port economics*. Routledge.
<https://www.routledge.com/Port-Economics/Talley/p/book/9780415771634>
- Yescombe, E. R., & Farquharson, E. (2018). *Principles of project finance* (2nd ed.). Academic Press.
<https://www.sciencedirect.com/book/9780128120668/principles-of-project-finance>
- Gatti, S. (2018). *Project finance in theory and practice: Designing, structuring, and financing private and public projects* (2nd ed.). Academic Press.
<https://www.sciencedirect.com/book/9780128113622/project-finance-in-theory-and-practice>
- Brealey, R. A., Myers, S. C., & Allen, F. (2020). *Principles of corporate finance* (13th ed.). McGraw-Hill Education.
<https://www.mheducation.com/highered/product/principles-corporate-finance-brealey-myers/M9781260565553.html>
- Berk, J., & DeMarzo, P. (2020). *Corporate finance* (5th ed.). Pearson.
<https://www.pearson.com/en-us/subject-catalog/p/corporate-finance/P200000003023>
- Damodaran, A. (2012). *Investment valuation: Tools and techniques for determining the value of any asset* (3rd ed.). Wiley.
https://pages.stern.nyu.edu/~adamodar/New_Home_Page/valbook.htm
- Benninga, S. (2014). *Financial modeling* (4th ed.). MIT Press.
<https://mitpress.mit.edu/9780262027287/financial-modeling>
- Finnerty, J. D. (2013). *Project financing: Asset-based financial engineering* (3rd ed.). Wiley.
<https://www.wiley.com/en-us/Project+Financing%3A+Asset+Based+Financial+Engineering%2C+3rd+Edition-p-9781118412275>